Peter Ramsey

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Curriculum Vitae

Research Interests

Topics: Algebraic combinatorics, algebraic topology, & computational algebra.
Objects: Matroids, hyperplane arrangements, & generalizations.

Education

PhD Mathematics *(Expected) May 2027*
Louisiana State University – Baton Rouge, LA
Advisor – Dr. Christin Bibby

MS Mathematics *May 2023*
Louisiana State University – Baton Rouge, LA

BA Pure Mathematics & Sociology *May 2021*
Spring Hill College – Mobile, AL

Publications

Published

The Orlik–Solomon algebra of a locally geometric poset
with Christin Bibby.
Séminaire Lotharingien de Combinatoire, Proceedings of FPSAC 2026, 95B.48 (2026), 12 pp.
Peer reviewed extended abstract.

Submitted for publication

The Orlik–Solomon algebra of a locally geometric poset and cohomology of complex abelian arrangements
with Christin Bibby.

Conference Participation

Invited Talk: The Orlik–Solomon algebra of a locally geometric poset *October 2026*
Special Session on Algebraic and Geometric Aspects of Matroid Theory, AMS Fall Central Sectional Meeting, University of Minnesota-Duluth, Duluth, MN.

Invited Talk: The Orlik–Solomon algebra of a matroid prescheme *September 2026*
LSU Combinatorics Seminar, Louisiana State University, Baton Rouge, LA.

Poster Presentation: The Orlik–Solomon algebra of a locally geometric poset *July 2026*
38th International Conference on Formal Power Series and Algebraic Combinatorics, University of Washington, Seattle, WA.

Participant: Topology of arrangements with an eye to applications *September 2025*
International Research Conference, University of Pisa, Pisa, Italy.

Participant: Séminaire de Mathématiques Supérieures (SMS) 2025: An Introduction to Recent Trends in Commutative Algebra *June 2025*
SLMath Summer School, Fields Institute, Toronto, Canada.

Honors & Awards

Research Training Group (RTG) Graduate Fellow *May 2024 – August 2026*
Graduate fellowship awarded to dedicate time towards research.

Certificate of Teaching Excellence *May 2024, May 2023*
Awarded to select graduate teaching assistants in the mathematics department for outstanding teaching ability.

Teaching Experience

Instructor of Record: Differential and Integral Calculus *Fall 2026*
Responsible for developing lesson plans, preparing assignments, delivering class lectures, and assigning final grades.

Instructor of Record: College Algebra *Fall 2022 – Spring 2024*
Responsible for delivering class lectures, tutoring in the LSU Math Lab, and assigning final grades.

Grader *Fall 2021 – Spring 2024*
Graded for courses in linear algebra, multidimensional calculus, differential equations, probability, and applied algebra.

Professional Service

Graduate Mentor: Summer Math Workshop *May 2025*
Supervised and mentored a small group of undergraduate students, facilitating discussion as they learned about selected math topics.

Graduate Mentor: Directed Reading Program *Fall 2023*
Mentored an undergraduate student in chess board structures in graph theory.

Co-Chair: Geaux Mathematics Committee *Summer 2023*
Organized and moderated sessions for teaching, research, and mental health during incoming graduate student orientation.

Graduate Teaching Mentor: Communicating Mathematics *Spring 2023 – Spring 2024*
Supervised and mentored new graduate students on instructing college algebra in a classroom setting.

Facilitator: ALEKS Test Preparatory Camp *Summer 2022*
Responsible for preparing and delivering lectures and facilitating group work sessions for undergrads preparing for the ALEKS placement test in mathematics.